

Executive Insights Summary: Mumbai Retail Operations Analysis

Project: End-to-End Sales & Profitability Analysis

Tech Stack: Python (Pandas, Numpy), T-SQL (SQL Server), Power BI

Objective: To analyze localized retail transaction data across the Mumbai Metropolitan Region (MMR) to identify key revenue drivers, assess geographical profitability, and provide data-backed recommendations for operational efficiency.

1. Key Analytical Observations

After processing and modeling approximately 10,000 transactional records, three primary trends emerged:

- **The Q4 Festive Surge (Seasonality):** Time-series analysis indicates a significant and consistent spike in total revenue during October and November. This correlates strongly with regional Diwali purchasing behaviors, highlighting a critical window for maximum revenue capture.
- **Margin Discrepancies by Geography:** While raw order volume is distributed across the MMR, profitability is not. The **Western Suburbs** (Andheri, Bandra, etc.) emerged as the most financially efficient zone, yielding the highest profit margin at **14.55%**, closely followed by Thane & Navi Mumbai at 14.12%.
- **Tech & Furniture Dominate Revenue:** The sub-category breakdown reveals that "Phones" (0.35M) and "Chairs" (0.34M) are the absolute top-grossing products, heavily outpacing traditional office supplies.

2. Strategic Business Recommendations

Based on the data visualizations, I recommend the following operational adjustments to maximize ROI:

- **Supply Chain Pre-Allocation for Q4:** Given the predictable Q4 revenue spike, the supply chain must be optimized in advance. Inventory for top-grossing items (Phones and Chairs) should be pre-allocated to regional warehouses specifically in high-volume zones like Thane & Navi Mumbai by late September to prevent stockouts and logistics bottlenecks during the festive rush.
- **Targeted Marketing in High-Margin Zones:** Since the Western Suburbs generate the highest profit margins per transaction, marketing and customer acquisition budgets should be disproportionately funneled into this region. Acquiring a customer here yields a better return on investment than in the Eastern Suburbs (which sits at a lower 10.00% margin).
- **Strategic Product Bundling:** Technology (Phones) is driving massive revenue. To pull up the profit margins of lower-performing sub-categories, the business should implement cross-selling strategies. For example, bundling high-margin tech accessories or office storage units with every Phone or Chair purchase can increase the average order value (AOV).

3. Engineering Approach

- **Data Wrangling:** Utilized Python (Pandas) to clean raw data, handle null values, parse inconsistent datetime formats, and algorithmically map generic geographies to specific Mumbai commercial zones.
- **Data Modeling:** Loaded the cleaned dataset into SQL Server to run aggregate queries, ensuring data integrity before connecting to Power BI.
- **Visualization:** Built an interactive, DAX-powered dashboard utilizing a star-schema model to allow stakeholders to filter KPIs by Year, Region, and Category dynamically.